

OrthoHMM: HMM-Centered Group Inference With Phylogenetic Refinement

Condensed scientific draft, native-evidence revision of 6 October 2026. Not submission-ready. This revision preserves the [third 4 October text](#), its 35-page assembled review and the rc4 archive. Fresh native ablations, the two-cell QfO figure and functional-pair composition below are later additions, not retroactive members of those artifacts. Native inference and remaining publication work are incomplete; the progress ledger and actual scheduler state govern execution. The prior revision's chronology remains recorded below.

This evidence-integrated working revision preserves the [second 4 October text](#), its 34-page review and the rc3 archive. The newly integrated inventory, candidate-cap mechanism and retained stage-cost addenda are not retroactively part of those artifacts. It also preserves the [first 4 October text](#) and its rendered reviews; it does not relabel their historical resource statements or change the frozen method, benchmark scores or endpoints. The [extended manuscript](#) retains detailed methods, historical analyses, citations and audit records. This main text does not supersede frozen protocols or historical result manifests.

Abstract

Orthology inference requires balancing homolog recovery against separation of paralogs. We evaluated an HMM-centered pipeline with candidate-family expansion and phylogenetic refinement against established orthology tools. QfO and OrthoBench were the primary development-exposed benchmarks; a frozen YGOB evaluation assessed transfer to additional taxa. Phylogenetic OrthoHMM improved on its high-sensitivity configuration but did not consistently outperform full OrthoFinder. On OrthoBench, their group-recovery F1 values were 74.11% and 72.74%, respectively. On corrected QfO, OrthoFinder had higher VGNC, SwissTrees and TreeFam-A point estimates, whereas OrthoHMM had higher GO/EC similarity and FAS. A matched-recall simulation control supported an initial HMM-search contribution in a fixed graph procedure, not full-pipeline superiority. Family-disjoint generalization and uncertainty for several QfO endpoints remain unresolved. The completed shared-host matched-resource panel retains all 27 attempts, with 24 eligible measurements; it does not establish isolated comparative efficiency. The evidence supports a bounded contribution and an explicit precision-recall trade-off, not universal accuracy or efficiency claims. Fresh native ablations retain neutral downstream-profile effects and a SwissTrees precision-recall trade-off whose adjusted F1 interval includes zero. Higher native GO/EC means reflect scored-pair membership differences, not changed serialized similarities on shared pairs. Synthetic null scoring revealed composition-dependent behavior of the approximate significance filter. Heterogeneous-length simulations favored full OrthoFinder; supplied generating trees did not establish an accuracy gain, and stronger topology perturbations reduced F1 and recall in several conditions. A fixed-candidate generating gene-tree diagnostic localized most divergent-condition recall loss upstream of reconciliation; residual event-history traces identified satellite-constraint losses and duplication evidence absent from retained candidates.

Methods

The retained configurations are high-sensitivity OrthoHMM and the satellite_v2 phylogenetic pipeline. The original OrthoHMM preprint (Steenwyk et al. 2024) describes its lineage, not all subsequent implementation changes. The prospective method was frozen at 7f3a9e4, with BLOSUM62, E-value threshold 1e-4, Leiden CPM resolution 0.1 and seed 4 (Traag, Waltman, and van Eck 2019; Traag, Van Dooren, and Nesterov 2011). The phylogenetic configuration expands candidate families, infers gene and species trees, and applies positive-paralogy pair inference. Historical runs retain their actual revisions rather than inheriting this prospective pin. The [method diagram](#) distinguishes initial search, profile refinement, candidate expansion and phylogenetic inference.

Initial search uses banded local maximum-path scoring with match, insert and delete states, integer BLOSUM62 emissions and fixed transition costs. Raw scores receive an approximate significance filter before geometric-mean length normalization. MSA-derived profiles learn match emissions, while retaining uniform transition costs; strict expansion requires both a member-score threshold and an initial sequence-supported anchor. This is not demonstrated HMMER/phmmer equivalence or calibrated assignment confidence. The [numerical specification](#) records the recurrence, constants, profile construction and acceptance rules at the full frozen source revision. No scoring/default change accompanies it.

A [prespecified null-score protocol](#) examined that significance filter without fitting parameters. Three residue compositions and lengths 50/150/400 each used ten seeds and 1,000 independently drawn query-target pairs per seed. Each pair had a one-target database and was scored with full and width-64 bands: 90,000 independent pairs, not 180,000 independent observations. Five fixed cutoffs yielded 90 tail endpoints with Bonferroni-adjusted exact binomial intervals. These intervals describe the synthetic generator, not reference-family or orthology uncertainty.

Comparators were OrthoFinder 3.1.5 (David M. Emms et al. 2026a, 2026b), SonicParanoid 2.0.9 (Cosentino, Sriswasdi, and Iwasaki 2024), ProteinOrtho 6.3.6 (Klemm, Stadler, and Lechner 2023), FastOMA 0.3.5 (Majidian et al. 2025) and OrthoMCL 1.4 (Li, Stoeckert, and Roos 2003). These method references do not replace run-specific executable provenance. OrthoFinder's sequence-only MCL checkpoint was a diagnostic output, not a separately finalized phylogenetic analysis. FastOMA used a supplied OrthoFinder species tree. QfO inputs included native ortholog pairs, native post-clustering relations or group-derived cross-species pairs as appropriate; these are not interchangeable output semantics. The [generated comparison](#) reports each conversion and prediction count.

OrthoBench measures curated group recovery (David M. Emms and Kelly 2020). QfO (Quest for Orthologs consortium et al. 2016; Altenhoff et al. 2020) reports GO and EC similarity, VGNC, SwissTrees and TreeFam-A F1, and FAS separately. Their arithmetic mean is a project-defined secondary summary, not an official QfO F1. Three Kingdoms is supplementary BUSCO-reference recovery (Manni et al. 2021), not genome-wide orthology truth. QfO and OrthoBench were repeatedly inspected during development. A retained family-evidence inventory scanned 1,785 immutable JSON result blobs at Git revision f145fff1 and 114 separately hash-frozen local-only reports, including historical parameter sweeps. The exact 70 OrthoBench/18 retained SwissTrees identities were canonicalized; score blocks, named/list mentions and unresolved ordinal/synthetic labels remained separate. Every block retained its file, JSON pointer, count-vector digest and recorded partition. Reports were not counted as independent native experiments or evidence of causal tuning. [Inventory protocol](#). The [YGOB protocol](#), using the curated homology/synteny resource (Byrne and Wolfe 2005), froze evaluation before test-score inspection; overlap assessment limits its interpretation to novel-taxon transfer rather than family-disjoint validation.

Uncertainty analyses resample declared reference units and recompute the benchmark statistic within replicates. They retain prespecified multiplicity adjustments, negative findings and unavailable contrasts. Curated families are not automatically independent: shared evolutionary history and predictions joining families can violate exchangeability. Dependent protein pairs are not treated as independent observations to obtain narrower intervals.

The prespecified QfO parameter neighborhood retained seven arms: the frozen control and six one-at-a-time changes. CPM resolutions were 0.08/0.12 versus 0.1, candidate minimum normalization 0.024/0.036 versus 0.03, and minimum margin 1.2/1.8 versus 1.5. SwissTrees F1 was recomputed from macro precision and recall in each of 100,000 shared family-bootstrap draws, not averaged across family F1 values. The seed was 20260925; Bonferroni adjustment retained all 18 planned endpoints across six contrasts and three metrics. This is development-exposed local sensitivity, not independent method selection. [Frozen parameter protocol](#).

Two evolutionary simulation panels used Zombi and Pyvolve (Davín et al. 2019; Spielman and Wilke 2015), ten seeds and seven conditions each: baseline, divergence, duplication/loss turnover, their combination, missing proteins, uneven sampling and a taxon-count control. Event-derived cross-species truth was not replaced by ancestral-family membership. Fixed 300-residue sequences and heterogeneous ancestral-family lengths of 100-500 residues were analyzed separately; descendants retain family length, without within-family indels. Native admission, including finite comparator graph weights, preceded scoring. Within each panel, 20,000 paired successful-seed resamples recomputed mean seed-level metrics, with a 14-contrast adjustment for primary F1. Failures were not imputed; available-case means from different seed sets are not paired effects. [Simulation protocols and results](#).

On the 70 heterogeneous-length datasets, phylogenetic OrthoHMM and full OrthoFinder also used supplied generating trees and deterministic rooted nearest-neighbor interchanges at clade distances two and four. Generating trees are oracle diagnostics, not achievable end-to-end inference or guaranteed accuracy upper bounds. All 126 F1/precision/recall endpoints were exploratory, with 20,000 paired-seed resamples, seed 20260918 and Bonferroni adjustment. [Tree-control protocol](#).

The subsequent fixed-candidate gene-tree diagnostic retained all 70 OrthoHMM generating-species-tree cells. Only inferred, single-ancestral-family candidates received generating gene trees, either at their generating root or rerooted by the frozen minimum-duplication/loss procedure. Candidates, species trees, reconciliation and satellite-constraint policy were fixed; native pairs had to reproduce exactly before intervention. Bypasses and mixed-ancestry candidates retained native predictions. Descriptive ten-seed means and counts were reported without population intervals or independent-confirmation claims. The oracle changes branch lengths/support as well as topology/root, and is not an accuracy upper bound or a topology-only causal control. [Gene-tree diagnostic protocol](#).

A retained upstream trace followed all 163,527 true pairs through significant initial HMM hits, direct final graph edges, undirected connectivity, final candidates and native pairs. The saved graph precedes refinement/expansion; it is neither initial search evidence nor ortholog output. Missing significant hits do not distinguish prefilter rejection, score filtering, caps or ranking. Gene-to-seed membership inside merged candidates was not reconstructed from incomplete sidecars. Independent igraph/SciPy component and native-row checks covered all 70 cells. [Upstream trace protocol](#).

Finally, a complete post hoc cohort selected every generating-root candidate with within-candidate FP/FN after screening all 10,125 candidates. All cross-species pairs in those candidates, including correctly classified pairs, were traced against original simulator event tables and reconciled XML. Induced ancestor clades, species overlap, raw pair calls, root groups and native satellite-constraint filtering had to

reproduce original truth and oracle counts. An independent XML/Biopython reader recalculated fixed-rule and constraint decisions without importing the trace worker. No defaults or endpoints changed. [Residual trace protocol](#).

A post hoc reproducibility diagnostic compared accepted candidate-merge traces from the original OrthoBench satellite arm and three already completed native repeats by round and named source/target memberships. All common cluster indices and forward/reverse hit counts were checked. A controlled 19-gene fixture used the unchanged frozen candidate engine, four attachments per anchor per round and two rounds, varying only initial cluster order or one satellite's scores by one floating-point unit. Rejected candidates and the origin of historical numerical differences were not reconstructed. [Candidate-cap diagnostic](#).

Fresh Native Ablation Protocol

A separate frozen execution panel comprises six fresh OrthoBench identities and seven fresh corrected-QfO identities. P denotes downstream profile expansion, C candidate-family expansion and R reconciliation; initial sensitive HMM search is on in every cell. The retained P1/C0/R0 configuration is a separately reused historical baseline, not an additional fresh run. Native inference uses the frozen method with fresh output directories and does not reuse phylogenetic checkpoints. These identities are distinct from the completed 27-attempt scaling panel. Terminal scheduler, scientific-output and resource-accounting reviews remain separate gates; recovery of valid scientific outputs after a wrapper or measurement failure does not repair the original timing result. [Frozen native plan](#).

Fresh RefOG or SwissTrees count vectors and metadata must match retained family records exactly before their original paired intervals can be reused. OrthoBench retains all 12 planned simple effects and the 36-endpoint adjustment, with 20,000 original draws (seed 20260918). QfO retains all 14 planned contrasts and the 42-endpoint adjustment, with 100,000 original draws (seed 20260922). No new bootstrap draws, independent confirmation, family-disjoint validation or total-HMM-versus-non-HMM comparison follows from this binding. Reconciliation also changes the QfO output from group-clique relations to inferred ortholog pairs.

A read-only functional diagnostic binds six native raw GO/EC/FAS tables through their original scientific admissions. It preserves original means and denominators while comparing shared and exclusive scored pairs. Independent CSV/Decimal parsing and SQLite joins check the actual tables without importing the primary comparison parsers. These are composition and dependence diagnostics, not confidence intervals, corrected endpoints or another annotation-scoring run. [Diagnostic protocol](#).

Shared-Host Resource Measurement

The replacement resource panel uses the local x86 Threadripper host, bizon, not the historical DGX/ARM environment. The frozen panel comprises high-sensitivity OrthoHMM, satellite_v2 OrthoHMM with inferred phylogeny and full OrthoFinder 3.1.5 on nested sets of 4, 8 and 12 proteomes, with three rotated repeats per method and size (27 sequential attempts). The input counts are 73,266, 165,168 and 251,378 proteins. Scientific settings, input bytes and order remain frozen. Native tasks use the same 32-CPU affinity and 128-GiB ceiling; each non-exclusive Slurm allocation reserves 64 slots to accommodate measurement and reporting work. The selected affinity uses distinct physical cores, but allocation does not reserve the host against unrelated processes.

Under the user-authorized 3 October amendment, background analyses do not invalidate a run merely by competing for resources. Launch capacity, source and input identity, process attribution, measurement continuity and failure handling still apply. Process snapshots and pressure observations bracket the native interval and retain background CPU demand, memory pressure and I/O diagnostics. Snapshot coverage is not continuous isolation certification, and CPU pressure can exceed nominal bounds because of non-atomic accounting; pressure magnitudes are diagnostic rather than timing-eligibility thresholds.

The primary endpoints are native-command monotonic elapsed time, native task-subtree CPU-stat change including wrapper work, and native-step lifetime memory peak including the launcher. They exclude separate preparation, conversion and scoring; the peak is not pure algorithm RSS. Terminal scheduler, runtime, environment, resource and native-output evidence is reviewed before the next identity. All attempts and failures are retained without fastest-run selection, automatic retries or estimated overhead subtraction. Median/range summaries require all three eligible repeats; observed ranges are not confidence intervals. Shared-host distortion is unknown and potentially method dependent. Historical DGX and earlier shared-host timings are not pooled with this panel.

Timing measurements were collected on a shared Threadripper while other analyses were running. Competition for CPU, memory bandwidth and I/O may have affected elapsed times, with an unknown and potentially tool-dependent impact. These are observed shared-host timings, not estimates of isolated performance. The 6 October authorization does not require the DGX or a quiet window; safe capacity and valid accounting remain launch prerequisites. [Execution amendment](#), [endpoint and continuation contract](#).

Results

Accuracy Depends On The Endpoint

Phylogenetic OrthoHMM achieved 74.1061% OrthoBench F1 versus 72.7365% for full OrthoFinder. In the exploratory eight-method comparison, the F1 difference was +1.370 percentage points (21-endpoint adjusted interval [-7.281, 12.166]). The precision difference was +15.705 [1.175, 30.634] and recall difference -13.151 [-26.947, -0.609]. This is a precision-recall tradeoff, not established F1 superiority. The complete panel uses 100,000 paired RefOG draws and remains conditional on family exchangeability and development exposure. [All 21 contrasts](#). The [complete interval figure](#) shows F1, precision and recall on the same difference scale without ranking methods by their observed effects. A fresh installed full run reproduced all 59,770 OrthoHMM groups exactly; this establishes reproducibility, not independent accuracy or a controlled runtime comparison. [Full-run verification](#). A second full run using a separately reconstructed base Python and fresh inference/reader environments also reproduced all groups and all 70 family score records. Native pair, confidence, reconciliation-event and hierarchy files were byte-identical to the earlier run. This extends same-host environment reproduction, not biological validation or complete archive restoration. [Reconstructed-base verification](#). A third full run used restored local execution assets and separately acquired upstream inputs. Its independent scientific admission reproduced all 59,770 groups, all 70 family score records and four native TSVs exactly, without reusing phylogenetic checkpoints. This validates local archive-to-results reproduction, not cross-host/OS restoration, independent accuracy or controlled timing. [Restored-archive verification](#).

On corrected QfO, phylogenetic OrthoHMM versus full OrthoFinder scored 0.901690 versus 0.988546 on VGNC, 0.833513 versus 0.848413 on SwissTrees, and 0.614864 versus 0.791918 on TreeFam-A. OrthoHMM had higher GO similarity (0.490349 versus 0.469548), EC similarity (0.965650 versus 0.936130) and FAS (0.762993 versus 0.691422). These endpoints differ in reference scope and eligibility; they do not form a single accuracy ranking. [All eight methods](#).

An exploratory VGNC deletion diagnostic removed each of 16,844 reference blocks and its incident scored pairs in turn, across all eight methods. Every comparator retained a negative F1 difference against full OrthoFinder for every deletion. For phylogenetic OrthoHMM the range was [-8.7353, -8.6729] percentage points. These are sensitivity ranges, not confidence intervals: dependent deletions change the fixed scored table, and single-block sign stability does not establish population or joint-deletion robustness. Native inference and eligibility were not rerun.

For corrected SwissTrees, phylogenetic-minus-full-OrthoFinder F1 was -1.4900 percentage points, with a multiplicity-adjusted interval of [-8.7078, 7.2090]. Phylogenetic-minus-high-sensitivity OrthoHMM was +14.8015 points [4.9791, 26.6015]. These conditional estimates use 18 development-exposed families and a 24-endpoint adjustment; inclusion of zero is not equivalence. Candidate expansion also changes between OrthoHMM configurations, so their contrast does not isolate reconciliation. Protocol and contrast evidence.

All 56 method-pair GO/EC comparisons had identical six-decimal scores on shared scored pairs. Aggregate differences arose from different eligible pair sets and their denominators. Restricting evaluation to intersections would change the endpoint rather than resolve uncertainty. Historical raw hashes were verified for all 16 input tables. Scored-pair audit. The complete pair-composition figure displays the shared-denominator and exclusive-pair terms for all seven comparators against full OrthoFinder. Phylogenetic OrthoHMM’s shared pairs account for 90.12% of its GO scored set but 46.40% of OrthoFinder’s, and 86.59% versus 58.00% for EC. The original rounded-mean differences, +2.080130 and +2.952029 score points, are the sums of much larger opposing terms. These fractions describe eligible scored pairs, not proteome-wide coverage; the decomposition is arithmetic, not causal attribution or paired uncertainty.

Component Evidence Is Bounded

The corrected QfO factorial supported candidate-expansion-by-reconciliation interactions on SwissTrees F1, while all four profile-refinement F1 intervals included zero. Reconciliation increased precision and reduced recall. These results support conditional component effects, not proof that profile expansion universally improves orthology inference. Complete factorial.

In 35 matched-recall simulation datasets spanning seven conditions and five seeds, HMM-derived search evidence yielded mean downstream graph F1 of 83.6097%, versus 80.5180% for DIAMOND (Buchfink, Reuter, and Drost 2021). The adjusted paired seed-block interval for the +3.0917-point difference was [1.6782, 4.5313]. Profile expansion and phylogeny were off. This is a development-exposed fixed-graph comparison, not a comparison against OrthoFinder or a matched-effort result. Score rankings and hit identities differ, so the design does not isolate a causal mechanism. Matched-recall control.

Fresh Native OrthoBench Effects

All six fresh OrthoBench cells have admitted scientific scores over 70 RefOGs; 420 family records match the retained factorial exactly. Partition agreement is not asserted for every non-reference gene. P0/C0/R0 retains failed wrapper 22427 with scientific recovery, not successful timing. The two omitted native cells and their contrasts remain unavailable rather than imputed.

Native Cell	Group-Recovery F1 (%)	Precision (%)	Recall (%)
P0/C0/R0	69.7634	78.8686	62.5429

Native Cell	Group-Recovery F1 (%)	Precision (%)	Recall (%)
P0/C0/R1	72.7050	87.4948	62.1922
P0/C1/R0	66.6395	64.4883	68.9391
P0/C1/R1	73.4023	81.4862	66.7776
P1/C0/R1	73.3114	87.4586	63.1038
P1/C1/R0	67.2074	64.7001	69.9169

Native point estimates, conditional interval binding, independent family-record readback.

Six of 12 planned simple effects have matching native records. With C0/R1 fixed, profile expansion changes F1 by +0.6064 percentage points (adjusted interval [-1.2114, 4.5210]; family wins/ties/losses 4/61/5). With C1/R0 fixed, its effect is +0.5680 [-1.1132, 4.2906] (6/59/5). Both intervals include zero. These are downstream-profile effects with initial HMM search retained, not the total HMM contribution or proof of general benefit.

With P0 fixed, reconciliation changes F1 by +2.9416 [0.2542, 8.1443] at C0 and +6.7628 [1.3762, 14.6116] at C1. Candidate expansion changes F1 by -3.1239 [-10.6588, 3.2751] at R0 and +0.6973 [-2.4590, 4.8209] at R1, while increasing recall and reducing precision in both cases. These conditional development-exposed group-recovery contrasts retain family-exchangeability and percentile-coverage limitations. They neither complete a fresh eight-cell factorial nor establish superiority over full OrthoFinder. Group co-membership is not resolved pairwise-ortholog truth. Interpretation and exact endpoints.

Fresh Native QfO Trade-Off And Pair Composition

Two of seven fresh corrected-QfO cells have admitted accuracy; five remain unavailable in this dated snapshot. Both shown cells have P0/C0, with initial HMM search on. They are not the selected-default all-tool comparison above. The recovered R1 scientific result retains original failed 22437 timing, null resources and timing ineligibility. Successful conversion, assessment and scientific admission do not turn that failed timing into a valid measurement.

Endpoint	Statistic	P0/C0/R0	P0/C0/R1
VGNC	F1	0.66683353	0.89818458
SwissTrees	F1	0.68918396	0.78957434
TreeFam-A	F1	0.60540356	0.60250783
GO	Similarity	0.47211905	0.49025964
EC	Similarity	0.93211406	0.96770222
FAS	Sample mean	0.77494473	0.78500682

Admitted native endpoint snapshot. The corresponding project-defined secondary means are 0.69009982 and 0.75553924, not official QfO F1 or a single accuracy ranking. Submitted relations decrease from 9,009,082 group-derived clique pairs to 5,113,820 inferred pairs; the fractions of 984,137 input accessions appearing in predictions are 55.5258% and 55.1107%. Mapping loss is zero in both conversions. These are prediction coverage and output-semantics observations, not accuracy statistics.

Only the R effect at P0/C0 has exact matched SwissTrees family records in both fresh native cells; the other 13 planned contrasts remain unavailable. Reusing the original draws and all 42 adjusted endpoints gives F1 +10.0390 percentage points with adjusted interval [-4.6418, 24.4199], precision +30.5212 [14.3252, 46.5473] and recall -6.5334 [-18.0413, -0.0436]. F1 wins/ties/losses are 14/1/3. The F1 interval includes zero; precision rises while recall falls. Original count-based arithmetic differs slightly from native serialized endpoints and remains separately reported, not substituted. The 18 families are development-exposed, and exchangeability and percentile coverage are conditional assumptions. Guarded interval binding, independent arithmetic readback.

Native QfO Figure. The reviewed four-panel figure separates the three F1 endpoints from GO/EC/FAS, shows SwissTrees precision and recall, and retains the zero-crossing adjusted F1 effect. The small TreeFam-A marker overlap does not conceal its decrease in the table above. Values come directly from the admitted snapshot and guarded binding; exact original-Python 3.10 replay is separate from Python 3.12 rendering. This is a two-cell ablation, not an initial-HMM-off control, a complete fresh factorial or a timing figure. Figure source/readback/visual review.

Every R-on GO scored pair (78,607) occurs in the R-off set (145,142), and every R-on EC scored pair (116,929) occurs in R-off (186,098). All common serialized six-decimal scores agree exactly. The 66,535 GO and 69,169 EC R-off-only pairs have lower means, 0.4506870677 and 0.8719528955. Thus, at retained precision, the higher native functional means reflect scored-pair membership and denominator composition, not changed similarities on common pairs or proof of correct ortholog selection. Intersection-only means would change the endpoint.

FAS realizes 38,205 R-off and 252,451 R-on sampled pairs, with only 1,007 shared pairs: 2.6358% of R-off and 0.3989% of R-on. All shared serialized scores agree; the original sample-mean difference remains +0.0100620873, not replaced by a shared-only difference of zero. Unseeded method-specific sample mixtures, missing scores and repeated proteins prevent this overlap from supplying an admitted paired confidence interval. Independent SQLite readback checks all 817,432 rows in the six bound tables. Neither membership arithmetic nor native pair-IID standard errors resolve other-endpoint comparison uncertainty. Functional composition result, independent SQL readback. No new score, default, independent-validation claim or timing repair accompanies this diagnostic. GO/EC/FAS, VGNC, TreeFam-A and secondary-mean paired uncertainty remains unfinished.

Synthetic Null Tails Depend On Composition

At the frozen $E < 1e-4$ cutoff, width-64 scoring passed 0/1/0 of 10,000 ordinary-background pairs at lengths 50/150/400. These sparse counts do not establish rare-tail calibration. Half-background-plus-half-glutamine sampling passed 89.83%, 100% and 100%; its adjusted intervals were [88.7484%, 90.8453%] and [99.9181%, 100%] for each longer length. All 90 endpoints and their figure are retained. A hit here is only a forced pair passing the approximate significance gate, not a predicted ortholog or an observed pipeline false positive. No prefilter or biological inference ran. The strong synthetic composition is not a measured real-proteome prevalence. The audit reproduces all endpoint counts and 180 sparse reference-Python scores, not all native scores. No coefficients, thresholds or defaults were fitted or promoted.

Evolutionary Simulations Retain Failures And Tree Sensitivity

In the fixed-length stress panel, high-sensitivity and phylogenetic OrthoHMM had 70 and 64 admitted datasets out of 70. No OrthoFinder output passed its native-completion/finite-graph gate; all 14 planned comparisons are unavailable, not wins for OrthoHMM. The heterogeneous-length panel admitted 70 and 67 OrthoHMM datasets and 65 full OrthoFinder datasets. Full OrthoFinder had higher paired mean F1

than both OrthoHMM modes in all seven conditions. Adjusted F1 intervals were below zero for all seven high-sensitivity contrasts and four phylogenetic contrasts. Phylogenetic-minus-full-OrthoFinder effects were -11.7433 points for divergence (five paired seeds) and -12.2842 for combined divergence/turnover (eight). These are success-conditioned comparisons, not failure-adjusted population estimates. Species-tree inference failures for OrthoHMM and nonfinite graph weights for OrthoFinder remain explicit. [Fixed stress results](#), [heterogeneous-length results](#).

The complete tree-control analysis retained 560 arm outcomes: 537 scored and 23 failed. No generating-minus-inferred adjusted interval excluded zero. The stronger perturbation reduced F1 and recall with adjusted intervals below zero in four OrthoHMM conditions and two OrthoFinder conditions: 12 endpoints; no precision interval excluded zero. Retained upstream artifacts agreed in 400 comparisons, differed in two and were unavailable in 18. The differing OrthoFinder cases prevent a strict tree-only causal attribution there. These deterministic perturbations and ten planned seeds do not establish robustness to arbitrary trees or empirical posterior uncertainty. [Complete tree results and figure](#).

Gene-Tree And Event-History Controls Localize Errors

Across 10,125 fixed candidates, 1,377 received eligible gene-tree controls; 8,748 bypasses remained unchanged. Among eligible controls, 300 differed in unrooted topology and 30 additional candidates only in root. Generating-root mean F1 changed by +0.136 to +0.734 percentage points across the seven conditions. These descriptive effects are not significance tests or evidence of independent improvement. In divergence and divergence/turnover, respectively 12,736 of 12,766 and 13,072 of 13,104 residual generating-root false negatives (99.765% and 99.756%) already cross candidate boundaries. Changing gene trees alone cannot recover them. [All-condition oracle results](#).

Of those cross-candidate losses, 11,442 and 11,554 (89.840% and 88.387%) lie in different saved graph components. The remaining 1,294 and 1,518 are connected but separated by candidates, including 117 and 128 direct graph edges. Almost all component-separated losses lack significant initial hits in either orientation. This identifies retained stage boundaries, not a single cause: missing hits cannot isolate the prefilter, and paths may cross other families. [Complete upstream trace](#).

The complete residual cohort contains 20 candidates: eight eligible controls and twelve bypasses. All 800 cross-species pair rows reproduce 380 TP, 292 TN, 62 FN and 66 FP. All 62 false negatives receive high-confidence speciation calls before unsupported satellite sources are detached; their root groups agree, but the post-constraint filter removes them. Of the false positives, 45 are bypass calls at true duplication ancestors and 21 receive speciation calls without retained species overlap. All 66 have overlap evidence in their full parent generating family that is absent from their retained candidate. Thus a single retained copy per species is not sufficient to establish orthology. These selected-cohort counts do not replace whole-panel accuracy. [Event-history residual results](#).

The trace supports candidate-retention and membership-filter explanations for these fixed simulation cases. It does not establish that disabling constraints, merging candidates indiscriminately, or changing a search threshold improves independent real-data inference. All negative and neutral findings, bypasses and upstream losses remain. QfO/OrthoBench retain their primary role; these privileged-information simulations are mechanistic support, not substitutes for curated-reference validation or unfinished biological error strata.

Local Parameter Sensitivity Did Not Establish Improvement

The complete seven-arm panel includes separately admitted private-runtime high-CPM recovery. Its SwissTrees F1 was 83.253016% versus 83.351322% for the frozen control, a difference of -0.098306 percentage points with adjusted interval [-1.050447, 0.747525]. All 18 adjusted intervals include zero; no statistically supported parameter improvement is established. Identical observed normalization-arm statistics do not imply identical whole-proteome predictions or equivalence on unseen families. No defaults were changed. The all-arm figure retains every contrast and the full adjustment denominator.

Historical SIGSEGV, allocator and admission failures remain failed. Recovery followed a validated content-equivalent private control and has separate inference, conversion and score evidence; it neither repairs those earlier attempts nor proves their cause or memory safety. This panel is not an OrthoFinder superiority test or a controlled runtime comparison.

Transfer And Biological Recovery Reveal Trade-Offs

Frozen YGOB group-recovery F1 was 92.233654% for phylogenetic OrthoHMM and 92.318524% for full OrthoFinder. Their difference was -0.084870 points, with adjusted interval [-0.622528, 0.445222]. OrthoHMM had higher precision and lower recall. The evaluation projects predictions onto the reference universe and assumes exchangeable pillars; it does not establish unrestricted generalization or resolved pairwise orthology. YGOB results.

A descriptive partition by the frozen overlap screen retained all original false-positive allocations. In screen-negative pillars, phylogenetic OrthoHMM versus full OrthoFinder had F1 of 57.93% versus 45.02%, precision of 55.92% versus 30.74%, and recall of 60.10% versus 84.03%. In screen-positive pillars, their F1 values were 93.37% versus 95.01%. The negative stratum contained 2,893 singleton pillars out of 3,298, compared with 2,017 out of 6,952 in the positive stratum. This composition difference and the screen's inability to exclude remote homology prevent independent-family or causal interpretations. The all-method stratum figure is descriptive, with no new uncertainty estimate or method tuning.

The prespecified whole-genome-duplicate application used experimental evidence from Kuzmin and colleagues (Kuzmin et al. 2020). All 240 experimental pairs were retained, with 239 input-eligible and 231 shared-reference-pillar pairs. On this development-exposed application, phylogenetic OrthoHMM separated 238 of 239 input-eligible pairs, versus 58 for high sensitivity and 236 for full OrthoFinder. Among the 231 reference-eligible pairs, however, only 193 OrthoHMM separations retained at least one non-*S. cerevisiae* reference homolog with each anchor, versus 56 for high sensitivity and 227 for full OrthoFinder. Mean per-pair homolog coverage was 82.338%, 99.149% and 98.413%, respectively. Coverage counts reference homologs in the union of the anchor groups and can be high even when the anchors are merged; it is not orthology recall. The complete five-method comparison also retains SonicParanoid and the diagnostic OrthoFinder MCL checkpoint.

Phylogenetic OrthoHMM minus full OrthoFinder had a supported-separation difference of -14.719 percentage points, with adjusted interval [-21.645, -8.225], and a coverage difference of -16.075 points [-19.755, -12.496]. These exploratory percentile intervals use 20,000 paired pillar resamples and a 12-endpoint adjustment. Each eligible pair occupies a distinct pillar; these conditional estimates do not establish independent generalization. Supported separation does not establish cross-species copy-specific orthology. In the six prespecified case traces, five focal homologs left their anchor groups during root-lineage reconstruction, before satellite constraints. They remained in other output groups; no example was replaced to improve the result.

A subsequent fixed-tree diagnostic evaluated all four existing root-duplication rules on the same seven candidate families and six examples, with its protocol frozen before alternative outcomes were examined. The supported-children and confidence rules reproduced the species-overlap baseline partitions exactly. The mapped-event rule reduced homolog coverage in four of five reference-eligible examples, without improving supported separation. None recovered the five focal homologs into either anchor group. Native pair predictions and confidence annotations remained unchanged. This post hoc intervention rejects these three alternatives as a repair on the fixed inputs, not as methods in general. It neither establishes topology error nor identifies ancestral-copy truth, and no defaults were changed.

Shared-Host Resource Panel

The snapshot retains 27 of 27 reviewed attempts, 25 with measured native resources and 24 eligible observations. 6 method/size cells have three eligible repeats.

These are shared-host matched-resource observations with 32 native CPUs and a 128-GiB limit per run, not isolated comparative timing. Contention distortion is unknown and potentially method dependent. No background overhead is subtracted.

Tables report median [minimum, maximum] only for cells with three eligible repeats. An unavailable summary is not zero or a failed native inference. Ranges are not confidence intervals. Eligibility counts retain measured exclusions and pre-native aborts rather than selecting the fastest attempts.

Native command wall (seconds)

Method	Proteomes	Eligible/planned	Median [minimum, maximum]
OrthoHMM high sensitivity	4	2/3	Unavailable
OrthoHMM high sensitivity	8	3/3	1238.9352 [1221.4444, 1303.6330]
OrthoHMM high sensitivity	12	3/3	2720.0774 [2621.4976, 3027.0605]
OrthoHMM inferred phylogeny	4	2/3	Unavailable
OrthoHMM inferred phylogeny	8	3/3	1999.5508 [1888.8847, 2086.9473]
OrthoHMM inferred phylogeny	12	3/3	4392.8716 [4042.4262, 4809.2924]
OrthoFinder 3.1.5 full	4	3/3	441.2890 [420.7400, 474.2999]
OrthoFinder 3.1.5 full	8	3/3	1139.9398 [1120.2548, 1179.9703]
OrthoFinder 3.1.5 full	12	2/3	Unavailable

Task-subtree CPU bracket (CPU-seconds)

Method	Proteomes	Eligible/planned	Median [minimum, maximum]
OrthoHMM high sensitivity	4	2/3	Unavailable
OrthoHMM high sensitivity	8	3/3	34395.7481 [33878.9946, 36610.5696]
OrthoHMM high sensitivity	12	3/3	78335.8306 [75126.8588, 86415.3192]
OrthoHMM inferred phylogeny	4	2/3	Unavailable
OrthoHMM inferred phylogeny	8	3/3	56008.3471 [52911.7819, 58240.7234]
OrthoHMM inferred phylogeny	12	3/3	125264.7375 [116000.9964, 136293.3322]
OrthoFinder 3.1.5 full	4	3/3	4602.7585 [4455.3737, 5008.8970]
OrthoFinder 3.1.5 full	8	3/3	17734.4088 [17232.2843, 18787.3118]
OrthoFinder 3.1.5 full	12	2/3	Unavailable

Native-step lifetime peak (GiB)

Method	Proteomes	Eligible/planned	Median [minimum, maximum]
OrthoHMM high sensitivity	4	2/3	Unavailable
OrthoHMM high sensitivity	8	3/3	7.724 [7.715, 7.932]
OrthoHMM high sensitivity	12	3/3	10.555 [10.533, 10.730]
OrthoHMM inferred phylogeny	4	2/3	Unavailable
OrthoHMM inferred phylogeny	8	3/3	7.681 [7.677, 7.702]
OrthoHMM inferred phylogeny	12	3/3	10.511 [10.503, 10.603]
OrthoFinder 3.1.5 full	4	3/3	6.775 [6.769, 6.928]
OrthoFinder 3.1.5 full	8	3/3	10.226 [10.215, 10.233]
OrthoFinder 3.1.5 full	12	2/3	Unavailable

Excluded attempt indices: 0, 17, 20. Pre-native abort indices: 17, 20; these have no native resource endpoints, not zero measurements. Unreviewed indices: none.

Across measured attempts, maximum observed foreign CPU demand ranges from 42.1424 to 72.8708 core equivalents. These are process-interval observations, not reservations or estimates of causal slowdown.

CPU includes the native task-subtree wrapper bracket. Peak memory includes the native-step launcher and is not pure algorithm RSS. Preparation, conversion and scoring are outside the native command timer. This single nested taxon series co-varies proteome count and taxon composition, not taxon-invariant scaling. Native output validation does not provide new prediction-accuracy evidence. Historical DGX and earlier shared-host timings are not pooled.

The first high-sensitivity attempt completed native inference but failed the original process-monitoring cadence criterion. It remains a measured exclusion; the prospectively tested scheduling repair does not convert it to a pass. Two further attempts aborted before native inference: an OrthoFinder/twelve-proteome repeat encountered a preparation hash race, and a phylogenetic OrthoHMM/four-proteome repeat exceeded the environmental-response deadline. Both retain null native endpoints and separately reviewed infrastructure repairs. None of these failures was retried to fill a cell or obtain faster timing. All 27 planned identities are terminal-reviewed, including exclusions; this is not 27 eligible measurements. The three incomplete cells have two eligible repeats each, not completed three-repeat summaries.

[Machine-readable final table](#), [all attempts](#), [complete-cell summaries](#), [final resource figure](#), [final terminal review](#), [generated section and provenance](#).

Retained Development Exposure And Candidate Reproducibility

The family-evidence inventory contains 134 OrthoBench score blocks across 84 files and 77 SwissTrees blocks across 16 files: 7,770 and 1,386 family-block associations respectively. All 88 canonical families have explicit score evidence. Each of the 35 originally validation-labelled RefOGs appears in 18 declared validation blocks; each original development RefOG appears in 28 development blocks, and all 70 appear in 21 all-partition blocks. The historical 35/35 split remains recorded chronology, not untouched publication validation. Distinct count vectors number 84 and 36; neither different nor equal vectors establish independence or duplicate execution. The inventory also retains 34 subset-ordinal candidate-diagnostic containers and two synthetic scored-label containers without misassigning biological identities. [Verified inventory](#).

Each original/native candidate trace contains 8,440 accepted merges. The three native repeats share 8,428, 8,439 and 8,426 semantic records with the original. Common cluster indices and hit counts agree; the maximum common support difference is $2.1316282072803006e-14$. Every common anchor with changed selected source memberships reaches the four-attachment cap in both traces. In the frozen 19-gene fixture, baseline, order-reversed and one-unit-perturbed cases leave satellite IDs 8, 0 and 7 unattached, respectively: one unattached satellite and eight accepted merges in every case. These IDs are not numbers of missing satellites. This demonstrates that capped selection and tied/near-tied evidence can change assignments without changing merge counts. It does not establish historical relabeling, score-bit provenance, gene-tree correctness or an accuracy advantage. [Complete diagnostic](#).

Historical Costs Remain Scope-Specific

A retained native-cost linkage checks six completed twelve-proteome Threadripper repeats against the original OrthoBench inputs and prescribed settings. All three high-sensitivity final partitions equal the original p1_c0_r0 partition. The first phylogenetic final partition matches the original p1_c1_r1; the other two have changed groups containing 149 and 172 genes. All three candidate partitions differ,

involving 70, 160 and 253 genes in affected groups. No changed group intersects a reference family. Affected-group gene counts are not counts of genes moved, and structural agreement is not new accuracy scoring or equivalence of all intermediate artifacts. Native wall medians are 45.335 and 73.215 minutes; these reuse the shared-host panel rather than provide new measurements or costs of the original cached executions. The patched native runtime differs from that deployment. [Native cost/partition linkage](#).

The selected corrected QfO high-sensitivity and phylogenetic rows bind to p1_c0_r0 group-derived pairs and p1_c1_r1 native pairs respectively. Their retained resource associations describe one shared cached-replay worker (2,987.81 seconds), two preparation arms (3.035759820 and 77.641008615 seconds) and native reconciliation (6,822.365357 seconds). The repeated replay value is one observation, not two repeats; process RSS, sampled tree RSS and lifetime cgroup peak retain different meanings. Neither stage sums nor these native configuration observations fill the sixteen unavailable original factorial full-pipeline costs. The selected rows' complete cached-execution costs remain unknown. [Exact QfO stage chains](#), [factorial stage costs](#).

The additive all-tool metadata register retains 24 method/dataset rows and all 72 scientific score positions, with explicit conversion semantics and historical resource supplements. Selected QfO stage associations are added without replacing prior intervals; table entries are not independent timed experiments. Input consumption, transitive runtime attestation and missing whole-pipeline costs remain separately identified rather than inferred. [All-tool metadata integration](#).

Discussion And Limitations

The supported contribution is an HMM-centered alternative with measurable component effects and explicit accuracy trade-offs. Initial HMM search has bounded support in matched-recall simulations; additional profile refinement has not demonstrated a general benefit. Neither simulation evidence nor an OrthoBench point advantage establishes superiority over full OrthoFinder.

The approximate significance formula is not universally calibrated under the tested synthetic conditions. The null experiment does not measure real-data orthology false-positive rates or causally explain any benchmark difference; independent calibration would be needed before claiming statistical significance for arbitrary compositions, search spaces, profiles or band settings.

Original TreeFam-A family mappings and complete source trees remain unavailable. Public QfO container searches now cover recognized benchmark build contexts for [all 40 listed tag names](#) at retained digests, without finding original NHX/mapping filename candidates. Other layers and archive-embedded or differently named contents remain outside that search; no family labels are inferred from its negative result. Public archive recovery yielded historical Selectome subtrees, but all are restricted to Euteleostomi. A [taxonomic coverage audit](#) found that 55,933 of 79,320 retained reference relations (70.52%) have at least one endpoint outside that clade. Those relations cannot be reconstructed from these subtrees under a species-consistent mapping; the other 23,387 relations are only potentially in scope, not demonstrated reconstructions. This is a coverage exclusion, not an accuracy effect or a basis for family-level intervals. VGNC dependence and rare-error diagnostics do not justify the candidate confidence-interval procedure. Valid paired uncertainty for GO/EC, FAS and the secondary mean is also unfinished. These gaps must remain visible alongside point estimates. The [corrected FAS sample audit](#) reproduces all eight means, but scored fractions range from 0.0067% to 58.46% of reported eligible pairs and every sample reuses proteins across pairs. Native pair-level standard errors do not resolve this comparison uncertainty. The [requested-score audit](#) found 1-49 missing new scores per method except OrthoMCL, which lacked 1,252 of 9,000. Assuming omitted scores lie in [0,1], its intended-sample mean is bounded by 0.724872-0.736975; this is not a confidence interval. A [native-complexity audit](#) found no above-limit

protein in saved new-score pairs, but found them in precomputed pairs. This agrees with the demonstrated cutoff mechanism without identifying historical omitted pairs or establishing comparison bias. The completed eligible-population audit matches all eight methods' native logged counts, reusing six fixed recounts and freshly scanning two. Pairs absent from the precomputed lookup constitute 5.13% of satellite_v2's eligible set, 16.86% of full OrthoFinder's and 82.05% of its sequence-only checkpoint's. Assuming their scores lie in [0,1] gives conservative full-mean bounds of 0.733966-0.785282 for satellite_v2 and 0.605946-0.774511 for full OrthoFinder. These overlapping bounds neither establish a full-eligible-set advantage nor replace native sample scores or confidence intervals. Some lookup-missing pairs were newly scored in native samples; those values are not used to tighten these bounds. An eight-method stratum-weight decomposition separates count rounding from omission-induced changes in the saved mixture. Holding the observed stratum means fixed, native-minus-reweighted-saved-strata differences are +0.001318947 FAS units for OrthoMCL and +0.000030716 for phylogenetic OrthoHMM. These are numerical diagnostics, not corrected benchmark scores or estimates of population bias. Omissions can also change the stratum means; the unknown omitted pair identities, unseeded selection and dependence remain unresolved. No FAS confidence interval or new method ranking follows. The completed simulations and tree perturbations do not cover arbitrary evolutionary conditions, and novel-taxon YGOB testing retains homolog-family overlap with development data. Generating-tree, upstream and residual diagnostics localize errors in the retained development-exposed panel but do not establish empirical gene-tree accuracy, ancestral-copy root-HOG truth or independent generalization. The development inventory establishes retained evaluation exposure, not a complete causal tuning history, all historical experiments or family-disjoint validation. Its frozen local input set must be supplied to reproduce that expanded inventory. Capped tie-sensitive candidate selection is a demonstrated reproducibility limitation, but does not justify benchmark-chosen score rounding or attributing changes to competing workloads. A method change would require a new freeze and independent confirmation. Candidate retention can remove species-overlap evidence, and strict satellite detachment can remove otherwise correct pair calls. The demonstrated cases motivate testable hypotheses, not unvalidated new defaults.

Recovered OrthoMCL results retain sequence-specific BLAST failures. The separately recovered high-CPM result does not replace failed historical runs; no missing score was imputed and no failure was recategorized as a success. Shared-host runs and the historical DGX panel are descriptive resource records. The replacement matched-resource Threadripper panel has completed under the shared-host amendment, not as an isolation-controlled experiment. Competing workloads may affect tools differently, and identical resource limits do not identify isolated speed or causal speedups. The final figure retains excluded measurements and missing eligible repeats; it cannot support an isolated efficiency ranking. Six cells have three eligible repeats, while three cells retain unavailable summaries. This single nested taxon series does not isolate taxon-count scaling from proteome composition. Whole-study executable/versioned archival packaging remains open, alongside redistribution limitations and journal-specific formatting. Completed resource reporting does not resolve the scientific uncertainty and validation gaps above.

Reproducibility And Availability

The extended manuscript links frozen protocols, source revisions, seeds, input/output manifests, conversion audits, executable workflows and generated figures. Installed inference and independent readback have been validated on full OrthoBench, but that is not proof of all-method cross-host portability. Historical locks are retained as provenance; patched replacement environments must be distinguished from the binaries used for original scores. The QfO parameter numerical component passed fresh local archive extraction, copied byte verification and reproduction of all 18 endpoints within 1e-12

using unchanged arithmetic. A Python-event guard rejected its original-checkout canary and observed no later original-path events or project-module imports. This is not OS containment, a new independent statistical implementation, raw-reference recount, native inference restoration, cross-host validation or redistribution clearance.

A separate descriptive-table component restored under standard-library Python 3.12 and reproduced all 208 rows and 984 score/difference cells across four SwissTrees tables and eight methods. This verifies arithmetic from retained sufficient statistics and annotations, not raw-source admission, new benchmark scoring or bootstrap uncertainty. An input-only private archive workflow then restored the complete duplication and fragment dependency panels, retaining all 11 and 1,774 original record occurrences respectively. Independently pinned archive and binding digests preserve source identities at new locations. All 27 affected regression/options cases passed using the restored inputs, including four raw-export regressions with unchanged assertions. The fresh children blocked original-checkout reads and subprocesses; these Python-event guards are not OS containment. Native annotation extraction/admission was not rerun, and private archives remain unuploaded with redistribution uncleared. Neither component establishes complete executable study restoration or all-method cross-host portability.

The null-score observations retain all 180,000 synthetic score evaluations and their source/runtime pins. Its public numerical readback checks all 90 tail endpoints without loading the native kernel; this is not a complete inference runtime archive.

The standalone YGOB arithmetic replay uses identifier-free retained counts to reproduce all 12 point metrics and six nominal/adjusted interval endpoints after local archive restoration. This is not native re-admission, proof of pillar exchangeability or independent family validation; original transfer scores and overlap limitations remain.

The standalone simulation replay reproduces retained-count arithmetic for the two length panels without historical file reads. It preserves fixed-length unavailable comparisons and variable-length conditional effects; native admission and tree-control inference are not rerun.

The final resource-reporting archive restores outside the repository and reproduces all three table files, PNG pixels and resource-section text exactly from its copied isolated reader. It includes all attempts and failures without opening original evidence paths. This is reporting portability, not raw-accounting or native reproduction, isolated timing, complete study restoration or redistribution clearance. The source/review export integration adds explicit result-helper and final-manuscript selection; fixture tests do not establish an actual full-study archive.

The progress ledger records completed work and unmet requirements. No submission-ready release or archival DOI is claimed. Dated main-text HTML/PDF and review archives preserve the exact inputs of their own snapshots; older versions are not rendered copies of revised Markdown. A refreshed review is not a complete executable study release. This revision incorporates the completed shared-host resource panel while also integrating the later simulation mechanism diagnostics and preserving scientific limitations. The three hash-pinned checked-summary reports support a standalone reporting replay using standard-library Python; it recomputes the finite-panel means, stage-count identities and residual classifications, not native search/inference or original XML admission. Its separate execution and package receipts govern what has actually been restored and reproduced. The second revision's 34-page review/rc3 archive and the third revision's 35-page review/rc4 archive remain historical. The latter includes the exposure, reproducibility and cost paragraphs, but neither archive includes this 6 October native-evidence revision. Fresh native scientific tables, bounded interval bindings, the reviewed QfO figure and functional-pair diagnostic retain their own source/admission identities and failure records. The new source revision requires its own render/review and eventual versioned bundle integration; those steps

have not been executed here. Existing native and statistical replay receipts remain valid only within their recorded scopes. No new endpoint selection, score, method default or independent-validation claim accompanies this revision.

References

The bibliography attributes the methods and resources cited above. Full execution evidence and additional dependency references remain in the extended manuscript; this selected list is not complete software attribution.

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